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1. SQL> show user;

```
SQL> show user;
USER is "SYS"
SQL>
```

Menampilkan user yang sedang aktf sekarang

2. SQL> show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
```

Menampilkan container yang sedang aktf sekarang

3. SQL> create tablespace ts1;

```
SQL> create tablespace ts1;
create tablespace ts1
*
ERROR at line 1:
ORA-01543: tablespace 'TS1' already exists
Help: https://docs.oracle.com/error-help/db/ora-01543/
```

Membuat tablespace bernama ts1 tapi error kaerna ada tablespaces bernama ts1.
Solusinya melakukan drop tablespace ts1 including contents and datafles;

```
SQL> drop tablespace ts1 including contents and datafiles;

Tablespace dropped.
```

4. SQL> SELECT DBMS_METADATA.GET_DDL('TABLESPACE','TS1') FROM dual;

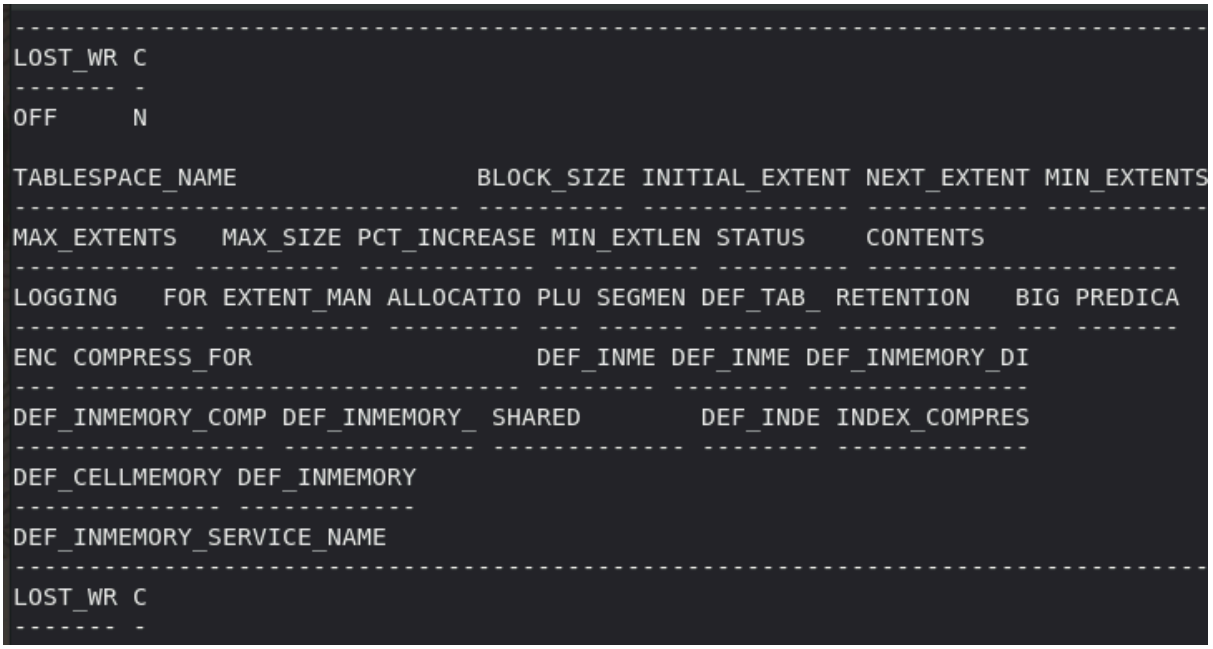
```
SQL> SELECT DBMS_METADATA.GET_DDL('TABLESPACE','TS1') FROM dual;

DBMS_METADATA.GET_DDL('TABLESPACE','TS1')
--------------------------------------------------------------------------------

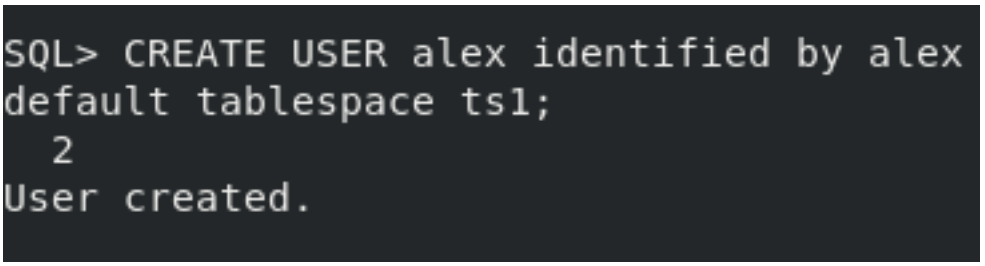
  CREATE BIGFILE TABLESPACE "TS1" DATAFILE
  SIZE 104857600
  AUTOEXTEND ON NE
```

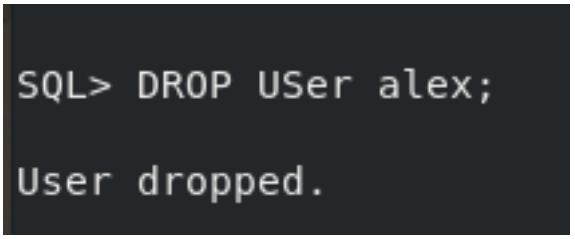
Menampilkan tablespaces yang bernama ts1

5. SQL> select * from dba_tablespaces
where tablespace_name='TS1' ;



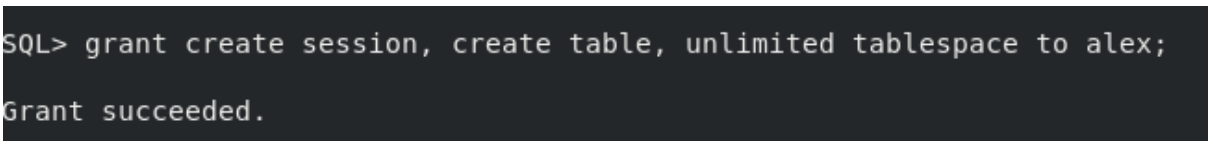
menampilkan seluruh isi data kolom dair table dba_tablespaces dimana tablescpaces
namanya adalah ts1
--REMEMBER THE RELATION
--TABLESPACE-SEGMENT-EXTENT-BLOCK
6. SQL> CREATE USER alex identfed by alex
default tablespace ts1;





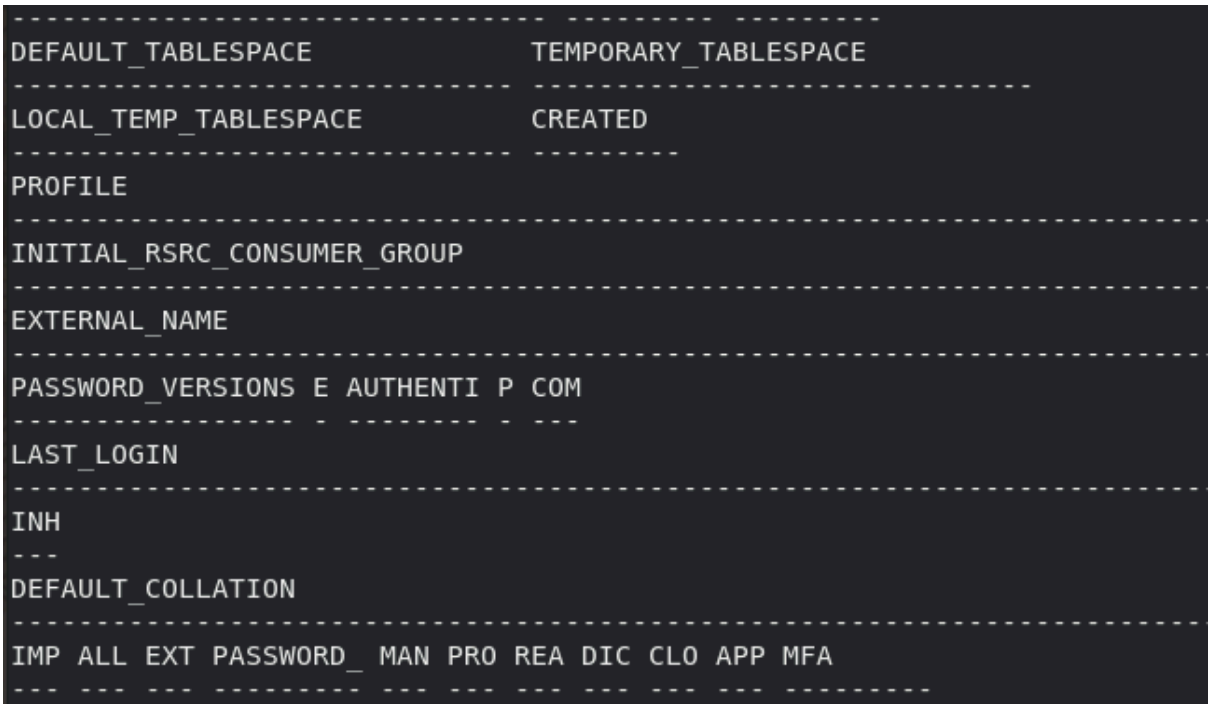
membuat user alex dan passwordnya alex dan memberikan default dari tablespace ts1. Tapi harus di drop dlu untuk user alex atau tdak error

7. SQL> grant create session, create table, unlimited tablespace to alex;

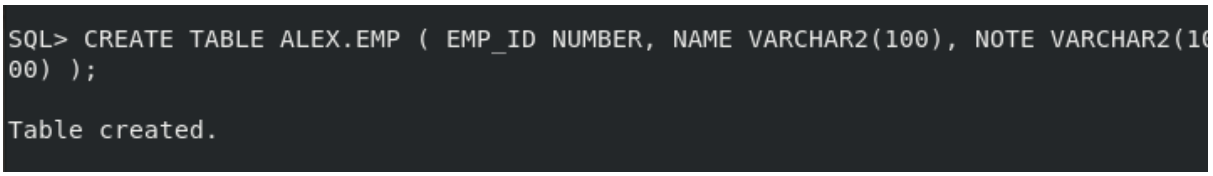


memberikan privillage ceate session, create table, unlimited tablespaces pada user

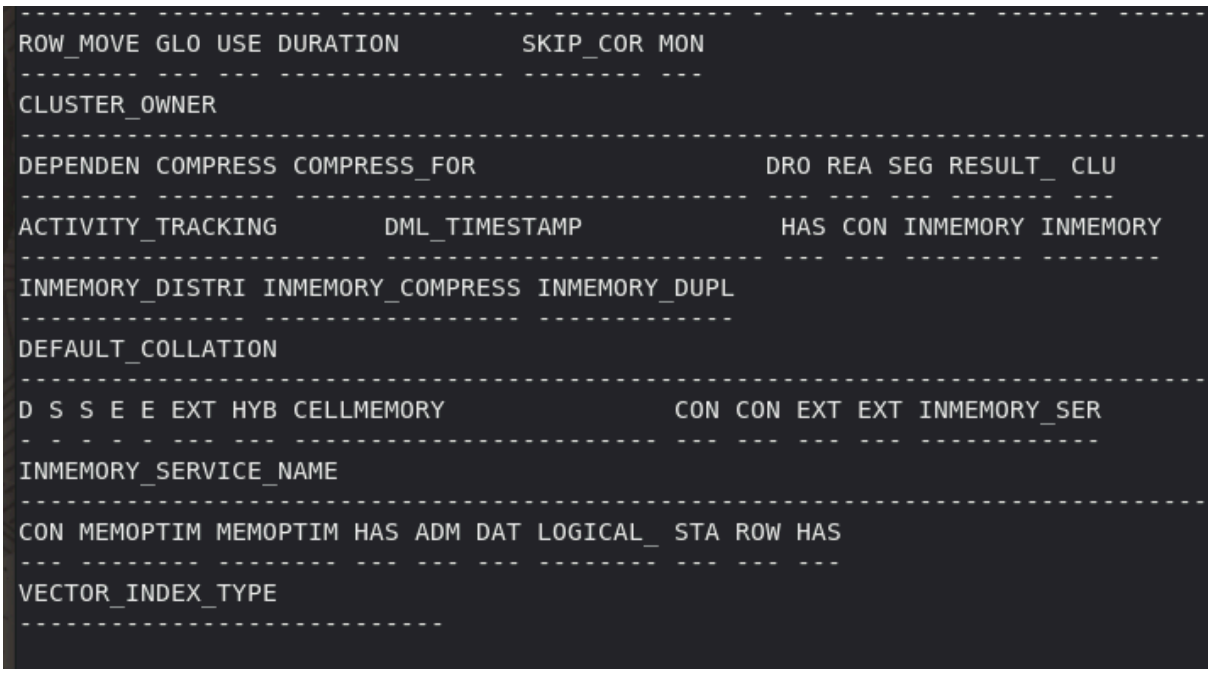
alex
8. SQL> select * from dba_users
where username='ALEX';



menampilkan seluruh isi kolom dari table dba_users
dimana username nya adalah ALEX
9. SQL> CREATE TABLE ALEX.EMP (EMP_ID NUMBER, NAME VARCHAR2(100), NOTE
VARCHAR2(1000))

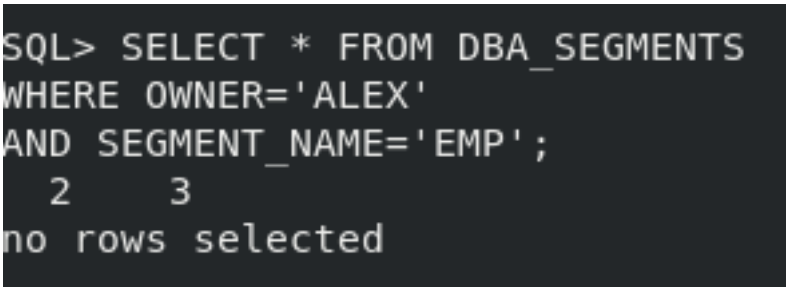


Membuat table emp dengan user nya alex dan beberapa atribute nya sepert emp_id dengan tpe data number, name dengan tpe data varchar2 dan maksimal karakter 100 dan varchar2 dari note dan maksimal karakter adlaah 10000
10. SQL> SELECT * FROM DBA_TABLES
WHERE TABLE_NAME='EMP'
AND OWNER='ALEX';



Menampilkan sleuruh isi data kolom dair table dba_tables

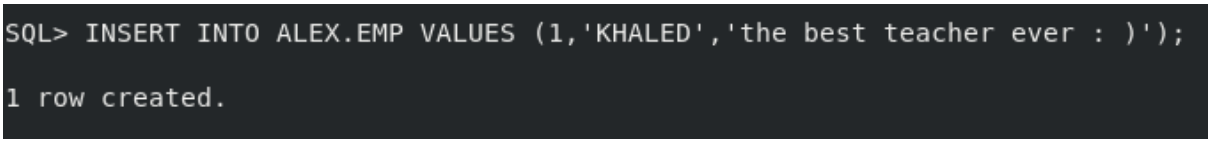
11. SQL> SELECT * FROM DBA_SEGMENTS
WHERE OWNER='ALEX'
AND SEGMENT_NAME='EMP';



--SO ORACLE DID NOT CREATE SEGMENT, NO ROWS IN THE TABLE

Menampilkan seluruh isi kolom dari table dba_segments dimana ownernya adalah alex, segment name nya EMP

12. SQL> INSERT INTO ALEX.EMP VALUES (1,'KHALED','the best teacher ever :)');



Memasukkan data pada table emp yang usenrya alex dengan isi baris 1 isinya adalah
khaleed dan the best teacher ever

13. SQL> commit;

```
SQL> commit;

Commit complete.
```

Menyimpan transaksi

14. SQL> SELECT * FROM DBA_SEGMENTS
WHERE OWNER='ALEX'
AND SEGMENT_NAME='EMP';

```
-------------- ------------ ------- ------- ------- -------- --------
INMEMORY_DISTRI INMEMORY_DUPL INMEMORY_COMPRESS CELLMEMORY
-------------- ------------ ----------------- ------------------------


OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                HEADER_FILE
------------------ ---------- ------------------------------ -----------
HEADER_BLOCK      BYTES     BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT
------------ ---------- ---------- ---------- -------------- -----------
MIN_EXTENTS MAX_EXTENTS   MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS
----------- ----------- ---------- ------- ------------ ------------ ----------
FREELIST_GROUPS RELATIVE_FNO BUFFER_ FLASH_C CELL_FL INMEMORY INMEMORY
--------------- ------------ ------- ------- ------- -------- --------
INMEMORY_DISTRI INMEMORY_DUPL INMEMORY_COMPRESS CELLMEMORY
-------------- ------------ ----------------- ------------------------
```

Menampilkan isi kolom dari dba_segments dimana owner nya adalah alexdan segment name nya adlaah EMP

15. SQL> select * from dba_extents
where owner='ALEX'
and SEGMENT_NAME='EMP';

```
--------------------------------------------------------------------------------
SEGMENT_TYPE       TABLESPACE_NAME                EXTENT_ID    FILE_ID
------------------ ------------------------------ ---------- ----------
  BLOCK_ID      BYTES     BLOCKS RELATIVE_FNO
---------- ---------- ---------- ------------
ALEX
EMP


OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       TABLESPACE_NAME                EXTENT_ID    FILE_ID
------------------ ------------------------------ ---------- ----------
  BLOCK_ID      BYTES     BLOCKS RELATIVE_FNO
---------- ---------- ---------- ------------
TABLE              TS1                                     0         27
       776      65536          8         1024
```

Menampilkan data kolom dari table dba_extents dimana ownernya adalah ALEX dan segment namenya adalah EMP

--now let us insert 1000 rows

16. SQL> begin
for i in 1..1000
loop
insert into ALEX.EMP values (i,'just test name', 'just test comments');
end loop;
commit;
end;

```
SQL> begin
 for i in 1..1000
 loop
 insert into ALEX.EMP values (i,'just test name', 'just test comments');
  2    3    4    5
  6  end loop;
  7  commit;
  8  end;
  9  /

PL/SQL procedure successfully completed.
```

memasukkan data tersebut sebanyak 1000 rows pada table EMP

17. SQL> SELECT * FROM DBA_SEGMENTS
WHERE OWNER='ALEX'
AND SEGMENT_NAME='EMP';

```
OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                HEADER_FILE
------------------ ---------- ------------------------------ -----------
HEADER_BLOCK      BYTES     BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT
------------ ---------- ---------- ---------- -------------- -----------
MIN_EXTENTS MAX_EXTENTS   MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS
----------- ----------- ---------- ------- ------------ ------------ ----------
FREELIST_GROUPS RELATIVE_FNO BUFFER_ FLASH_C CELL_FL INMEMORY INMEMORY
--------------- ------------ ------- ------- ------- -------- --------
INMEMORY_DISTRI INMEMORY_DUPL INMEMORY_COMPRESS CELLMEMORY
--------------- ------------ ----------------- ------------------------
```

Menampilkan isi kolom dari table dba_segments dimana ownernya adalah alex dan segment namenya adalah EMP

18. SQL> select * from dba_extents
where owner='ALEX'
and SEGMENT_NAME='EMP';

```
SEGMENT_TYPE       TABLESPACE_NAME                EXTENT_ID    FILE_ID
------------------ ------------------------------ ---------- ----------
  BLOCK_ID      BYTES     BLOCKS RELATIVE_FNO
---------- ---------- ---------- ------------
ALEX
EMP


OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       TABLESPACE_NAME                EXTENT_ID    FILE_ID
------------------ ------------------------------ ---------- ----------
  BLOCK_ID      BYTES     BLOCKS RELATIVE_FNO
---------- ---------- ---------- ------------
TABLE              TS1                                     1         27
       784      65536          8         1024
```

menampilkan isi kolom dari table dba_extents dimana ownernya adalah ALEX dan segment namenya adalah EMP

--now let us insert 1000000 rows

19. SQL> begin
for i in 1..1000000
loop
insert into ALEX.EMP values (i,'just test name', 'just test comments');
end loop;
commit;
end;

```
SQL> begin
 for i in 1..1000000
 loop
 insert into ALEX.EMP values (i,'just test name', 'just test comments');
  2    3    4    5
  6  end loop;
  7  commit;
  8  end;
  9  /

PL/SQL procedure successfully completed.
```

memasukkan data sebanyak 1000000 ke dalam table emp dari user alex

20. SQL> SELECT * FROM DBA_SEGMENTS
WHERE OWNER='ALEX'
AND SEGMENT_NAME='EMP';

```
OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                HEADER_FILE
------------------ ---------- ------------------------------ -----------
HEADER_BLOCK      BYTES     BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT
----------- ---------- ---------- ---------- -------------- -----------
MIN_EXTENTS MAX_EXTENTS   MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS
----------- ----------- ---------- ------- ------------ ------------ ----------
FREELIST_GROUPS RELATIVE_FNO BUFFER_ FLASH_C CELL_FL INMEMORY INMEMORY
--------------- ------------ ------- ------- ------- -------- --------
INMEMORY_DISTRI INMEMORY_DUPL INMEMORY_COMPRESS CELLMEMORY
--------------- ------------- ------------------ ----------------------
```

Menampilkan isi kolom dari dba_segments dimana owner nya adalah alex dan segment name nya EMP

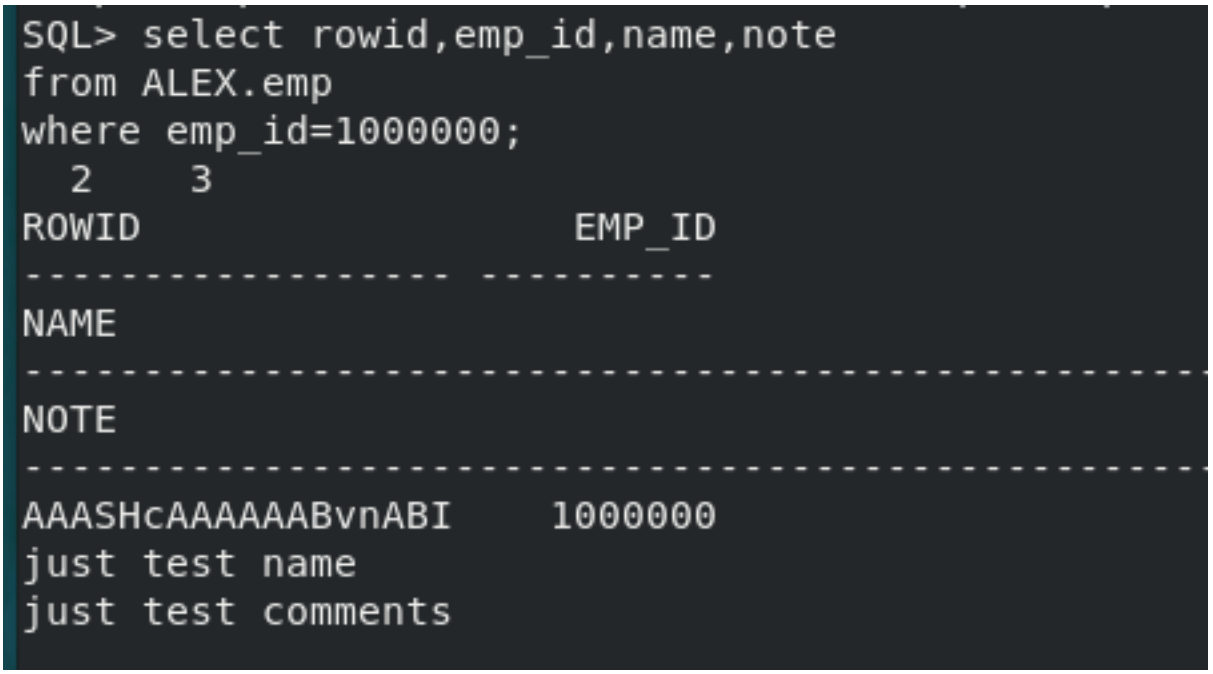
21. SQL> select * from dba_extents
where owner='ALEX'
and SEGMENT_NAME='EMP';

```
OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       TABLESPACE_NAME                EXTENT_ID    FILE_ID
------------------ ------------------------------ ---------- ----------
  BLOCK_ID      BYTES     BLOCKS RELATIVE_FNO
---------- ---------- ---------- ------------
TABLE              TS1                                    63         27
      7040    1048576        128         1024


64 rows selected.
```

Menampilkan isi kolom dari table dba_extents dimana ownernya adalah ALEX dan segment namenya adalah EMP

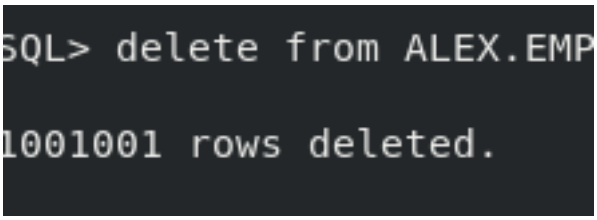
22. SQL> select rowid,emp_id,name,note
from ALEX.emp
where emp_id=1000000;



menampilkan rowid, emp_id, name, note dari table emp dan user dari ALEX dimana emp_id nya adalah 1000000

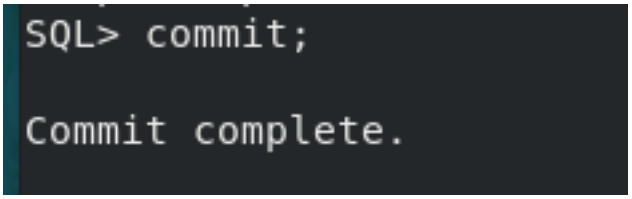
--example
--rowid OOOOOOFFFBBBBBBRRR
--OOOOOO object number
--FFF Relatve File Number
--BBBBBB The data block number
--RRR The row number

23. SQL> delete from ALEX.EMP;



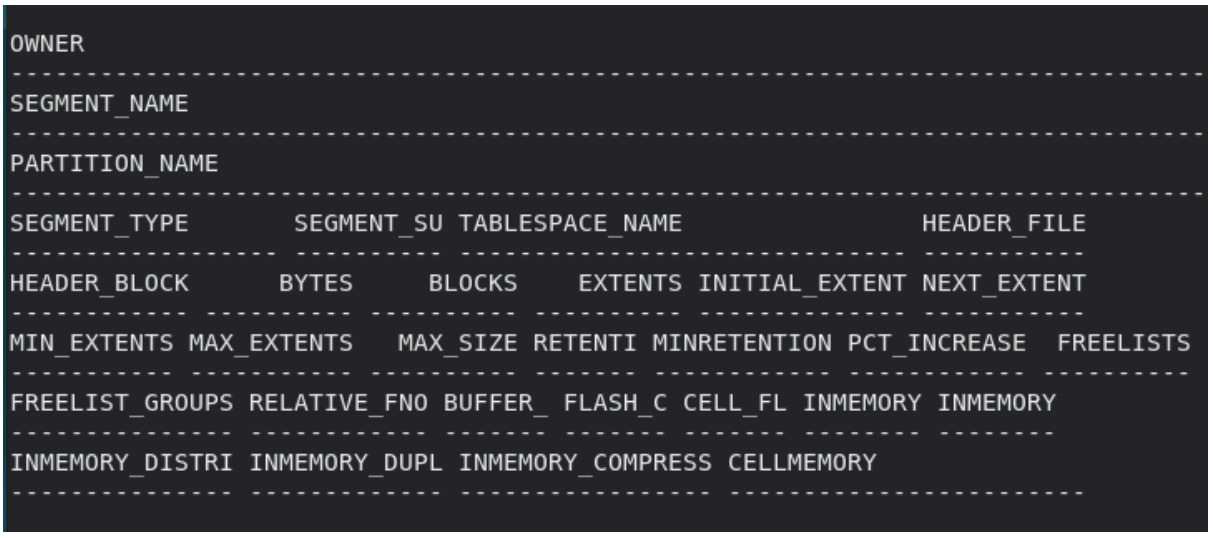
Menghapus table emp dari user alex

24. SQL> commit;



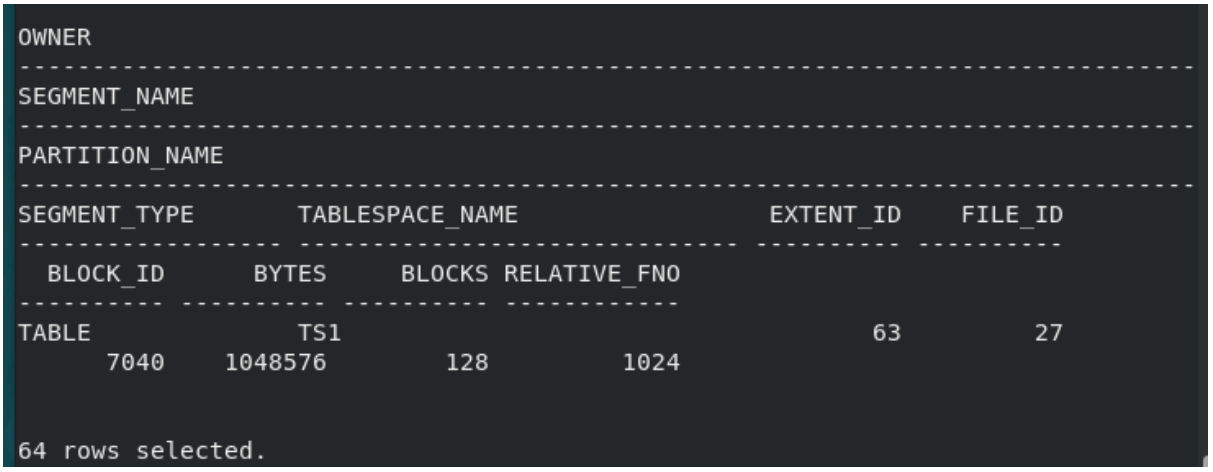
Menyimpan transaksi

25. SQL> SELECT * FROM DBA_SEGMENTS
WHERE OWNER='ALEX'
AND SEGMENT_NAME='EMP';



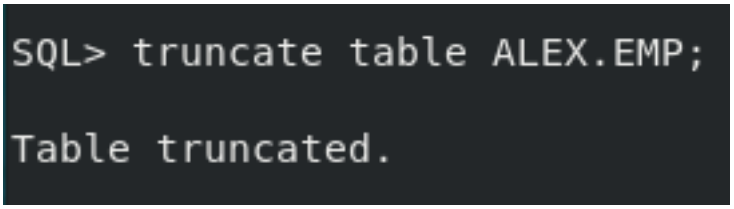
Menampilkan isi kolom dari table dba_segments dimaan ownernya adalah ALEX dan segment namenya adalah EMP

26. SQL> select * from dba_extents
where owner='ALEX'
and SEGMENT_NAME='EMP';



Menampilkan isi kolom dari table dba_extents dimana owner nya adalah ALEX dan segment namenya EMP

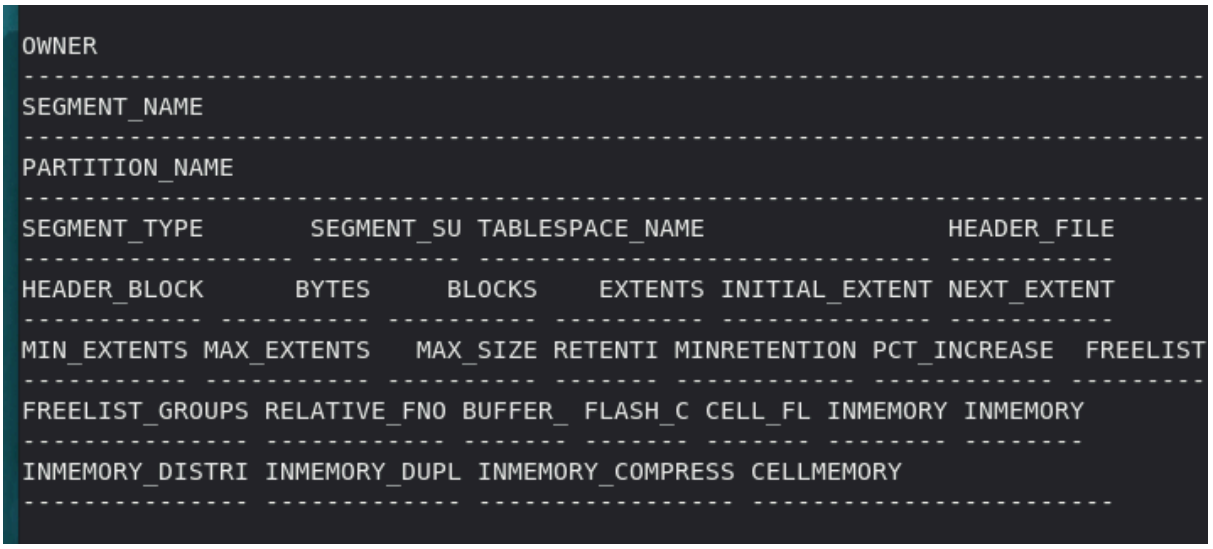
27. SQL> truncate table ALEX.EMP;



Menghapus seluruh isi table dari emp

28. SQL> SELECT * FROM DBA_SEGMENTS
WHERE OWNER='ALEX'

AND SEGMENT_NAME='EMP';



Menampilkan isi dari dba_segments dimana ownernya adalah ALEX dan segment namenya adalah EMP

29. SQL> select * from dba_extents
where owner='ALEX'
and SEGMENT_NAME='EMP';



Menampilkan isi kolom dari table dba_extents dimana ownernya adalah ALEX dan segment namenya adalah EMP

==========
==========

30. SQL> show user;
31. SQL> show con_name;

```
SQL> show user;
USER is "SYS"
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
```

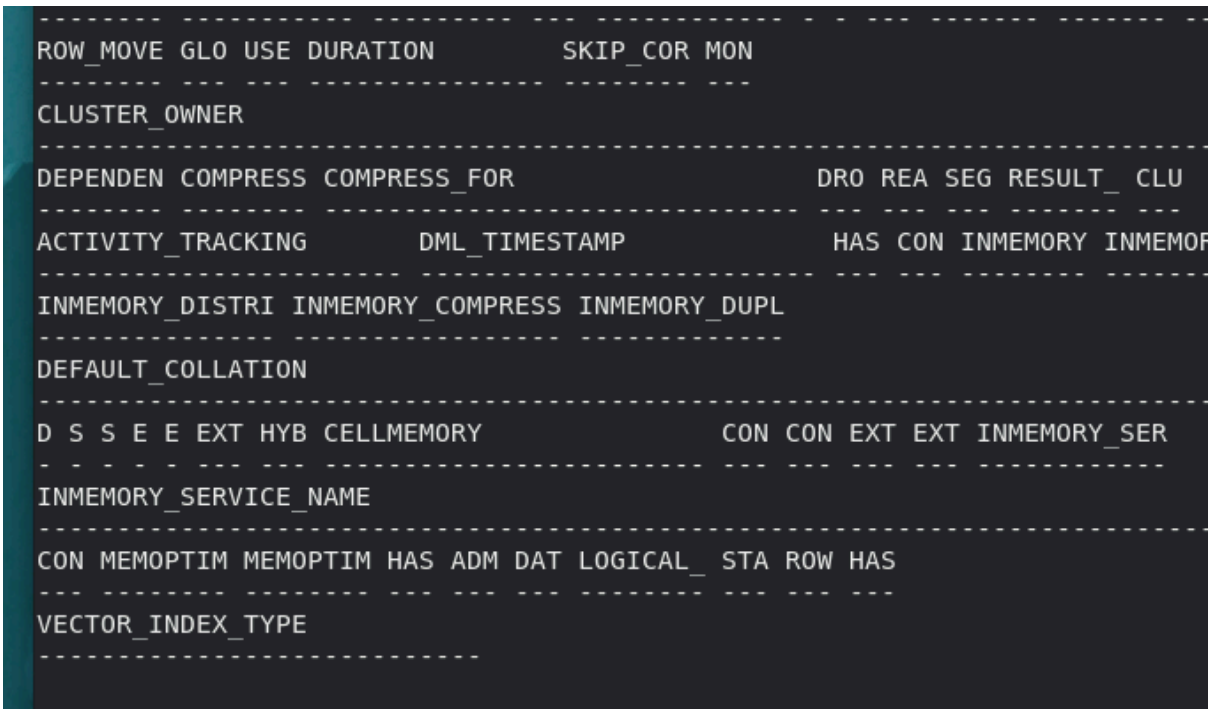
Menampilkan user dan container name yang sedang aktf sekarang

32. SQL> create table dept
(deptno NUMBER(2),
dept_name varchar2(100)
);

```
SQL> create table dept
(deptno NUMBER(2),
 dept_name varchar2(100 )
 );
  2    3    4
Table created.
```

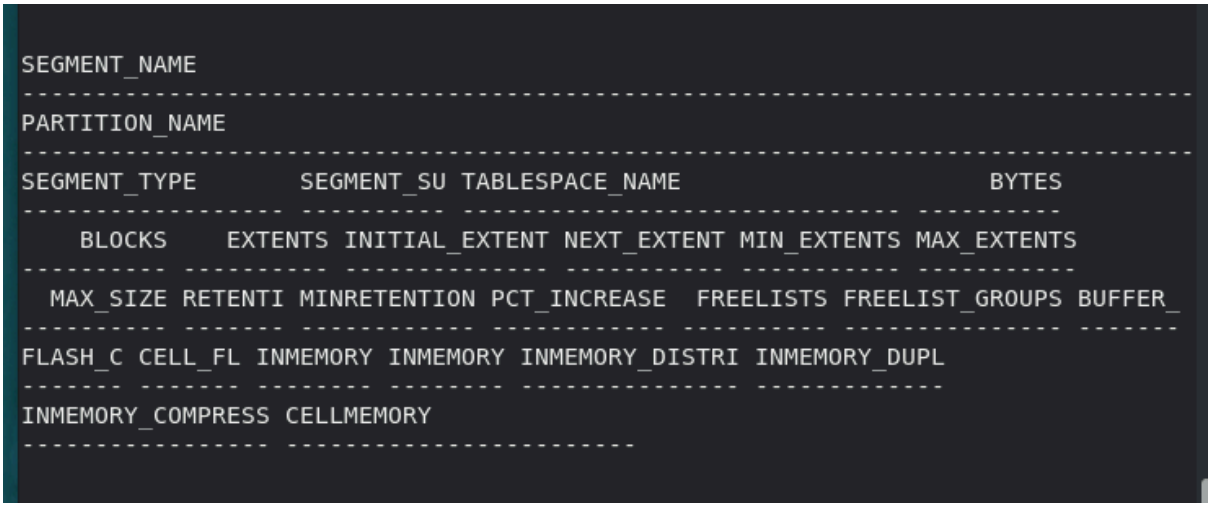
Membuat table baru bernama dept dengan atribute deptno tpenya number dan maksimal karakter 2, dept name dengn tpe varchar2 dengan tpe maksimal karakter adalah 100. Tapi disini error karena table dept sudah ada, kita harus melakukan drop table terlebih dahulu

33. SQL> select * from dba_tables
where table_name='DEPT';



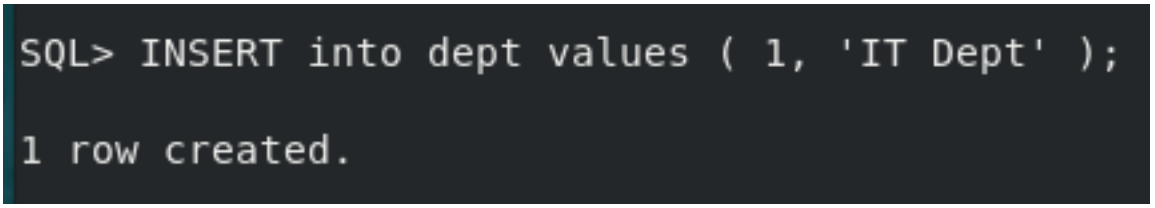
Memanmpilkan isi kolom dari table dba_tables dimana table namenya adalah DEPT

34. SQL> SELECT * FROM USER_SEGMENTS
WHERE SEGMENT_NAME='DEPT';



Menampilkan isi kolom dari table user_segments dimana segment namenya adalah DEPT

35. SQL> INSERT into dept values (1, 'IT Dept');



Memasukkan data dengan isi 1 dan IT Dept

36. SQL> INSERT into dept values (2, 'FINANCE');

```
SQL> INSERT into dept values ( 2, 'FINANCE' );

1 row created.
```

Menambahkan data pada table dept dengan isi baris 2 berisi fnance

37. SQL> commit;

```
SQL> commit;

Commit complete.
```

Menyimpan transaksi

38. SQL> SELECT * FROM USER_SEGMENTS
WHERE SEGMENT_NAME='DEPT';

```
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                     BYTES
------------------ ---------- ------------------------------ ----------
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
----------------- ------------------------
```

Menampilkan isi kolom dari table user_segments dimana segment name nya adalah DEPT

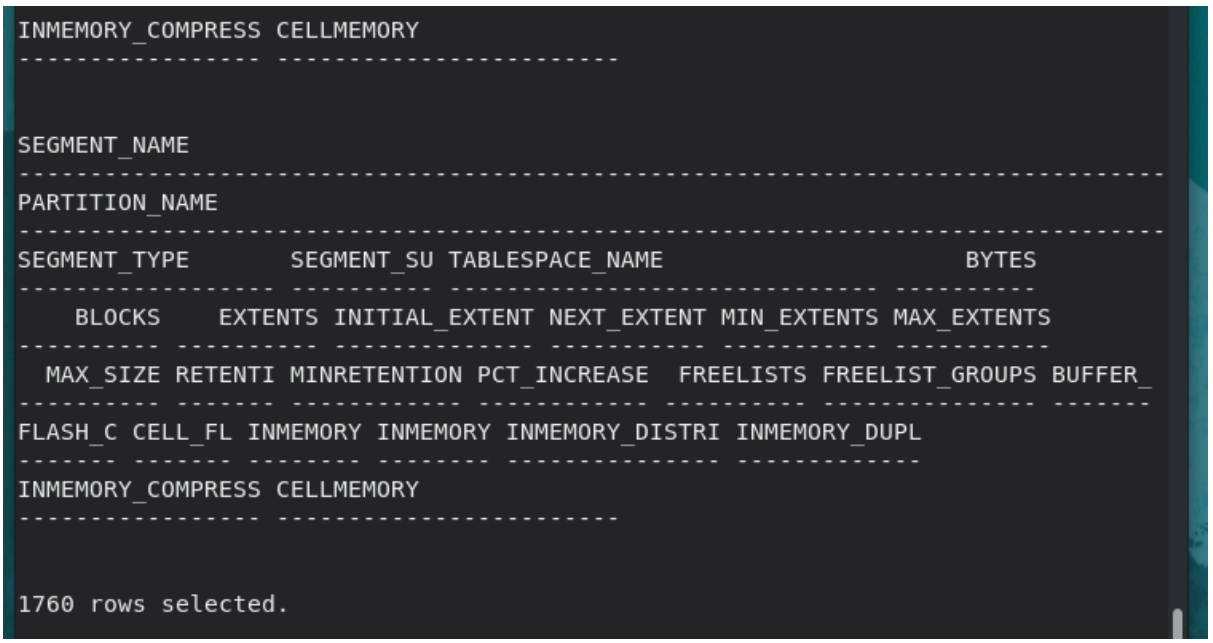
39. SQL> create unique index dept_inx on DEPT(deptno);

```
SQL> create unique index dept_inx on DEPT(deptno);

Index created.
```

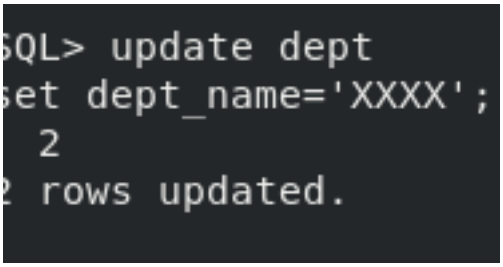
Membuat unique indext dept_inx pada table dept dengna atribute deptno

40. SQL> SELECT * FROM USER_SEGMENTS;



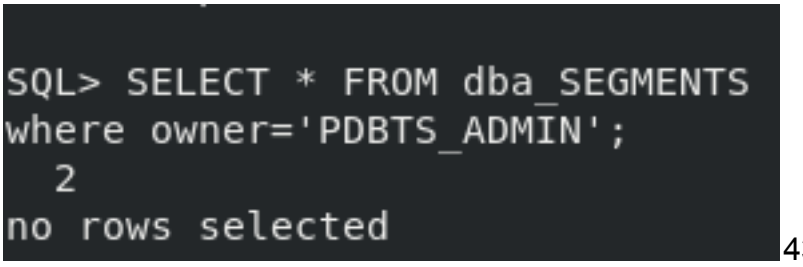
Menampilkan isi kolom dari table user_segments

41. SQL> update dept
set dept_name='XXXX';



melakukan update pada table dept dengan mengubah dept_name nya menjadi XXXX

42. SQL> SELECT * FROM dba_SEGMENTS
where owner='PDBTS_ADMIN';



43. SQL> select a.sid,
a.serial#, a.username, b.used_urec, b.used_ublk
from v$session a,
v$transacton b
where a.saddr = b.ses_addr
order by b.used_ublk desc;

```
SQL> select a.sid, a.serial#, a.username, b.used_urec, b.used_ublk
from   v$session a,
       v$transaction b
where  a.saddr = b.ses_addr
order by b.used_ublk desc;
  2    3    4    5
       SID    SERIAL#
---------- ----------
USERNAME
--------------------------------------------------------------------------------
 USED_UREC  USED_UBLK
---------- ----------
        39      19690
SYS
         2          1
```

--USED_UBLK = Number of undo blocks used
--USED_UREC = Number of undo records used
Menampilkan view session yang didefnisikan sebagai a, yang dimana mena,pilkan sid, serial#, username dan menampilkan juga view transascton yang di defnisikan sebagai b, yang ditampilkan adalah used_urec, used_ublk dimana a.saddr adalah b.ses_addr dan diurutkan dari b.used_ublk desc;
44. SQL> ROLLBACK;

```
SQL> ROLLBACK;

Rollback complete.
```

Melakukan rollback dimana untuk mengembalikan proses yang sudah terlanjur kita leewat sebelumnya
45. SQL> select a.sid, a.serial#, a.username, b.used_urec, b.used_ublk
from v$session a,
v$transacton b
where a.saddr = b.ses_addr
order by b.used_ublk desc;

```
SQL> select a.sid, a.serial#, a.username, b.used_urec, b.used_ublk
from   v$session a,
       v$transaction b
where  a.saddr = b.ses_addr
order by b.used_ublk desc;
  2    3    4    5
no rows selected
```

Menampilkan kembali sid, serial, username dari view session a, menampilkan used_urec, used_ublk dari view transacton b setelah kita melakukan rollback dimana saddr dari view sess = ses_addr diurutkan dari used_ublk descending

==========
==========
--DEFERRED_SEGMENT_CREATION
46. SQL> show user;
47. SQL> show con_name;

```
SQL> show user;
USER is "SYS"
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
```

Menampilkan user dan container name yang sedang aktf sekarang
48. SQL> show parameter DEFERRED_SEGMENT_CREATION;

```
SQL> show parameter DEFERRED_SEGMENT_CREATION;

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------
deferred_segment_creation            boolean     TRUE
```

Menampilkan parameter dari deferred segment creaton
--this mean the segment will created for the table
--1 afer you crate the table using DDL statement (create table)
--2 afer you insert one record (DML)
49. SQL> create table country (id number, name varchar2(100));

```
SQL> create table country (id number, name varchar2(100) );
create table country (id number, name varchar2(100) )
             *
ERROR at line 1:
ORA-00955: name is already used by an existing object
Help: https://docs.oracle.com/error-help/db/ora-00955/
```

```
SQL> drop table country
  2  ;

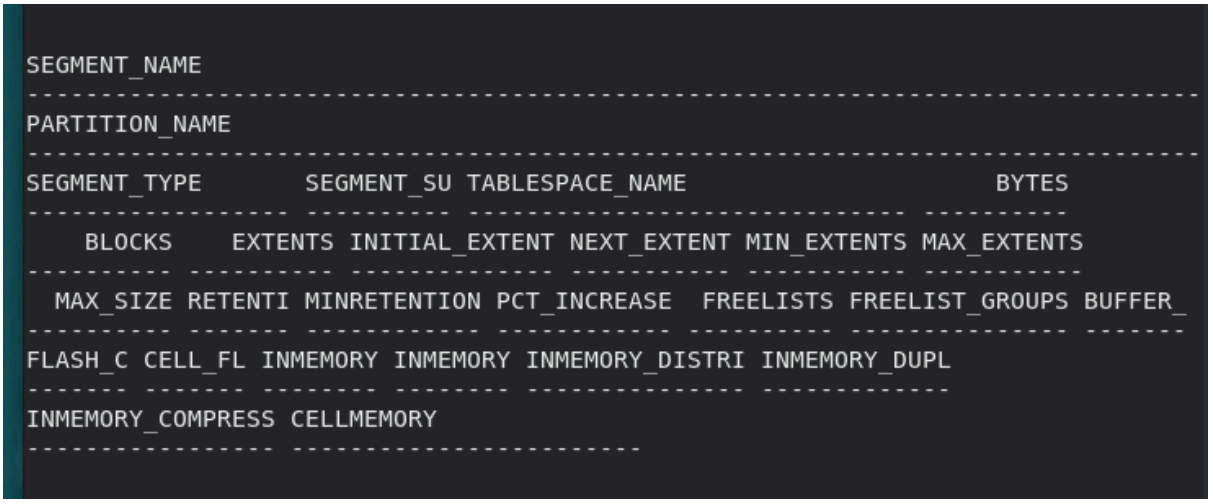
Table dropped.

SQL> create table country (id number, name varchar2(100) );

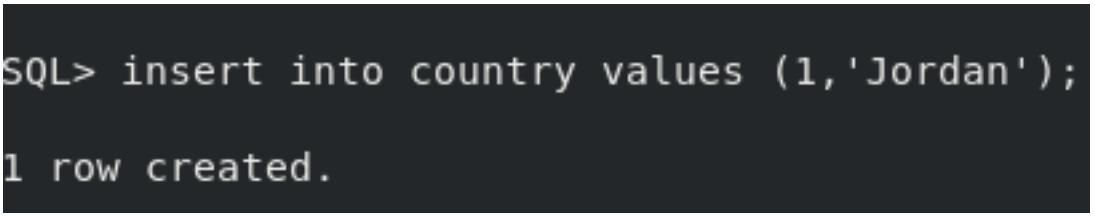
Table created.
```

membuat table country yang dimana terdapat atribute id dengan tpe number, name

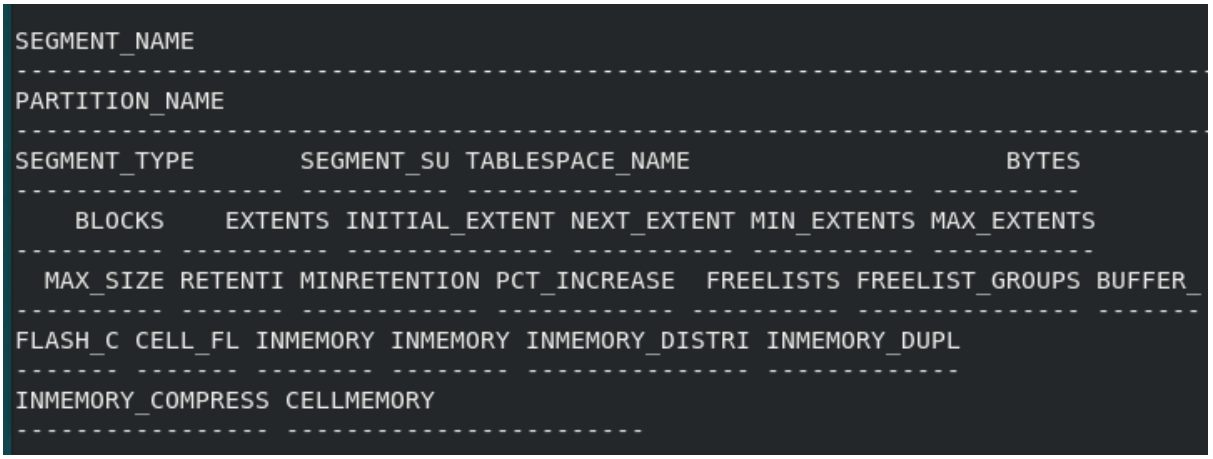
dengan tpe varchar2 dan maksimal karakter adalah 100. Tapi error karena table country itu sudah ada dan kita harus melakukan drop table terlebih dahulu
50. SQL> select * from user_segments
where segment_name= upper('country');



Menampilkan isi kolom dari table user_segments dimaan segment namenya adalah COUNTRY
51. SQL> insert into country values (1,'Jordan');



Menampilkan isi dari kolom pertama yaitu 1 dan isi nya adalah jordan dari table country
52. SQL> select * from user_segments
where segment_name= upper('country');



Menampilkan isi kolom dari table user_segments dimana segment_namenya adalah COUNTRY
--now let change DEFERRED_SEGMENT_CREATION to be false
--you can do this by alter session / alter system

--this will create the segment immediate
53. SQL> alter session set DEFERRED_SEGMENT_CREATION=false;

```
SQL> alter session set DEFERRED_SEGMENT_CREATION=false;

Session altered.
```

Mengubah session deferred segment creaton menjadi false
54. SQL> show parameter DEFERRED_SEGMENT_CREATION

```
SQL> alter session set DEFERRED_SEGMENT_CREATION=false;

Session altered.

SQL> show parameter DEFERRED_SEGMENT_CREATION;

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------
deferred_segment_creation            boolean     FALSE
```

Menampilkan parameter dari defered segment creaton
55. SQL> create table jobs (id number, ttle varchar2(100));

```
SQL> create table jobs (id number, title varchar2(100) );
create table jobs (id number, title varchar2(100) )
            *
ERROR at line 1:
ORA-00955: name is already used by an existing object
Help: https://docs.oracle.com/error-help/db/ora-00955/


SQL> drop table jobs;

Table dropped.
```

Menamabhkan table jobs dengan atribute id tpe numbeer, ttle dengan varchar2(100). Tapi disiini error karena nama table tersebut sudah ada yaa ?
56. SQL> select * from user_segments
where segment_name= upper('jobs');

```
SQL> select * from user_segments
where segment_name= upper('jobs');
  2
no rows selected
```

Menampilkan isi kolom dari table user segments dimana segment namenya adalah JOBS
--note

--Deferred segment is the default for tables, indexes, and parttons
57. SQL> alter session set DEFERRED_SEGMENT_CREATION=true;

```
SQL> alter session set DEFERRED_SEGMENT_CREATION=true;

Session altered.
```

Mengubah session dimana deferred segment creaton adalah true
--but we can do this also
58. SQL> create table test (n number)
SEGMENT CREATION IMMEDIATE;

```
SQL> create table test (n number)
  2  SEGMENT CREATION IMMEDIATE;

Table created.
```

Membuat table test dimana terdapat atribute n tpe nya number, dimana mengalokasikan penyimpanan dari table test
59. SQL> select * from user_segments
where segment_name= upper('test');

```
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
----------------- ------------------------


SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                    BYTES
------------------ ---------- ------------------------------ ----------
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
----------------- ------------------------
```

Menampilkan isi kolom dari table user_segments dimana segment name nya adalah TEST
60. SQL> create table test10 (n number)
SEGMENT CREATION DEFERRED;

```
SQL> create table test10 ( n number )
SEGMENT CREATION DEFERRED;
  2  create table test10 ( n number )
*
ERROR at line 1:
ORA-14224: Deferred segment creation is not supported for this schema
Help: https://docs.oracle.com/error-help/db/ora-14224/
```

Membuat table test10 dimana terdapat atribute nya adalah n dan tpenya number, tapi error karena statusnya sudah deferred

61. SQL> select * from user_segments
where segment_name= upper('test10');

```
SQL> select * from user_segments
where segment_name= upper('test10');
  2
no rows selected
```

Menampilkan isi kolom dari table user_segments dimana segment namenya merupakan TEST10

==========
==========

--Space-Saving Features (UNUSABLE index)

62. SQL> show user;
63. SQL> show con_name;

```
SQL> show user;
USER is "SYS"
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
SQL>
```

Menampilkan user dan container name yang aktf sekarang

64. SQL> show parameter DEFERRED_SEGMENT_CREATION;

```
SQL> show parameter DEFERRED_SEGMENT_CREATION;

NAME                                 TYPE        VALUE
------------------------------------ ----------- ----------
deferred_segment_creation            boolean     TRUE
SQL>
```

Menampilkan parameter dari deferred segment creaton

65. SQL> create table employees
(emp_id number,
fname varchar2(100),
lname varchar2(100),
constraint employees_pk primary key (emp_id)
);

```
SQL> create table employees
( emp_id number,
  fname varchar2(100),
  lname varchar2(100),
 constraint employees_pk primary key (emp_id)
 );
  2    3    4    5    6
Table created.
```

Membuat table employees dimana terdapat atribute yaitu emp_id dengan tpe number, fname dengan tpe varchar2 dan maksimal karakter 100, lname dengan tpe varchar2 dan maksimal karakter 100, dan menetapkan emp_id adalah pk nya

66. SQL> select * from user_tables
where table_name='EMPLOYEES';

```
-------- ---------- -------- --- ----------- - - --- ------- ------- -------
ROW_MOVE GLO USE DURATION        SKIP_COR MON
-------- --- --- --------------- -------- ---
CLUSTER_OWNER
--------------------------------------------------------------------------------
DEPENDEN COMPRESS COMPRESS_FOR                  DRO REA SEG RESULT_ CLU
-------- -------- ------------------------------ --- --- --- ------- ---
ACTIVITY_TRACKING       DML_TIMESTAMP             HAS CON INMEMORY INMEMORY
----------------------- ------------------------- --- --- -------- --------
INMEMORY_DISTRI INMEMORY_COMPRESS INMEMORY_DUPL
--------------- ----------------- -------------
DEFAULT_COLLATION
--------------------------------------------------------------------------------
D S S E E EXT HYB CELLMEMORY               CON CON EXT EXT INMEMORY_SER
- - - - - --- --- ------------------------ --- --- --- --- ------------
INMEMORY_SERVICE_NAME
--------------------------------------------------------------------------------
CON MEMOPTIM MEMOPTIM HAS ADM DAT LOGICAL_ STA ROW HAS
--- -------- -------- --- --- --- -------- --- --- ---
VECTOR_INDEX_TYPE
--------------------------
```

Menampilkan isi kolom dari table user_tables dimana EMPLOYEES merupakan table
namenya

67. SQL> SELECT * FROM USER_INDEXES;

```
AVG_DATA_BLOCKS_PER_KEY CLUSTERING_FACTOR STATUS     NUM_ROWS SAMPLE_SIZE
----------------------- ----------------- -------- ---------- -----------
LAST_ANAL DEGREE
--------- ----------------------------------------
INSTANCES                                PAR T G S BUFFER_ FLASH_C CELL_FL USE
---------------------------------------- --- - - - ------- ------- ------- ---
DURATION        PCT_DIRECT_ACCESS
--------------- -----------------
ITYP_OWNER
--------------------------------------------------------------------------------
ITYP_NAME
--------------------------------------------------------------------------------
PARAMETERS
--------------------------------------------------------------------------------
GLO DOMIDX_STATU DOMIDX FUNCIDX_ JOI IOT DRO VISIBILIT DOMIDX_MANAGEM SEG ORP
--- ------------ ------ -------- --- --- --- --------- ------------- --- ---
INDEXIN AUT CON
------ --- ---


2351 rows selected.
```

Menampilkan isi kolom dari table user_indexes

68. SQL> select * from user_segments
where segment_name IN ('EMPLOYEES','EMPLOYEES_PK');

```
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                     BYTES
------------------ ---------- ------------------------------ ----------
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
----------------- ------------------------
```

Menampilkan isi kolom dari table user_segments dimana segment name nya yang cocok dengan table employees dan employees_pk (yang kemungkinan jadi PK indeks)

69. SQL> insert into employees values (1,'khaled','alkhudari');

```
SQL> insert into employees values (1,'khaled','alkhudari');

1 row created.
```

Menambahkan value kepada table employees dimana baris pertama berisi khaled alkhudari

70. SQL> commit;

```
SQL> commit
  2  ;

Commit complete.
```

Melakukan penyimpanan transaksi

71. SQL> select * from employees;

```
SQL> select * from employees;

    EMP_ID
----------
FNAME
--------------------------------------------------------------------
LNAME
--------------------------------------------------------------------
         1
khaled
alkhudari
```

Menampilkan isi kolom dari table employees

72. SQL> select * from user_segments
where segment_name IN ('EMPLOYEES','EMPLOYEES_PK');

```
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                    BYTES
------------------ ---------- ------------------------------ ----------
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
----------------- -----------------------
```

Menampilkan isi kolom dari table user_segments dimana segment name nya cocok dengan table employees dan employees_PK

73. SQL> create index fname_index on EMPLOYEES(fname) ;

```
SQL> create index fname_index on EMPLOYEES(fname) ;

Index created.
```

Membuat indeks bernama fname_index pada table employees, menggunakan kolom

fname
74. SQL> select * from user_segments
where segment_name IN ('EMPLOYEES','EMPLOYEES_PK' ,'FNAME_INDEX');

```
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                     BYTES
------------------ ---------- ------------------------------ ----------
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
----------------- ------------------------
```

Menampilkan isi kolom dari table user_segments dimana segment name cocok denagn employees, employees_pk, fname_index
75. SQL> create index lname_index on EMPLOYEES(lname) UNUSABLE ;

```
SQL> SQL> create index lname_index on EMPLOYEES(lname) UNUSABLE ;

Index created.
```

Membuat indeks baru bernama lname_index pada table employees menggunakan kolom lname, namun kita menandai database untuk indeks ini tdak bisa digunakan
76. SQL> select * from user_segments
where segment_name IN ('EMPLOYEES','EMPLOYEES_PK'
,'FNAME_INDEX','LNAME_INDEX');

```
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                     BYTES
------------------ ---------- ------------------------------ ----------
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
----------------- ------------------------
```

Menampilkan isi kolom dari table user_segments dimana segment namenya cocok dengan employees, employees_pk, fname_index, lname_index
77. SQL> alter index LNAME_INDEX REBUILD;

```
SQL> alter index LNAME_INDEX REBUILD;

Index altered.
```

Mengubah indeks dari lname_index agar aktf atau bisa digunakan

78. SQL> select * from user_segments
where segment_name IN ('EMPLOYEES','EMPLOYEES_PK'
,'FNAME_INDEX','LNAME_INDEX');

```
--------- ------- ----------- ----------- ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
---------------- ------------------------


SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                     BYTES
------------------ ---------- ------------------------------ ----------
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
---------------- ------------------------
```

Menampilkan isi kolom dari table user_segments dimana segment_name nya cocok dengan employees, employees_pk, fname_index, lname_index

==========
==========

--basic compression
--With basic compression, the Oracle Database server compresses data at the tme of
performing bulk load using
--operatons such as direct loads or CREATE TABLE AS SELECT
--Is recommended for bulk loading data warehouses

79. SQL> show user;
80. SQL> show con_name;

```
SQL> show user;
USER is "SYS"
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
```

Menampilkan user dan contaner name yang aktf sekarang

81. SQL> select * from dba_objects
where rownum<10000;

```
OWNER
--------------------------------------------------------------------------------
OBJECT_NAME
--------------------------------------------------------------------------------
SUBOBJECT_NAME
--------------------------------------------------------------------------------
 OBJECT_ID DATA_OBJECT_ID OBJECT_TYPE             CREATED   LAST_DDL_
---------- -------------- ----------------------- --------- ---------
TIMESTAMP           STATUS  T G S  NAMESPACE
------------------- ------- - - - ----------
EDITION_NAME
--------------------------------------------------------------------------------
SHARING            E O A
------------------ - - -
DEFAULT_COLLATION
--------------------------------------------------------------------------------
O S I S CREATED_APPID CREATED_VSNID MODIFIED_APPID MODIFIED_VSNID
- - - - ------------ ------------ ------------- -------------


9999 rows selected.
```

Menampilkan isi kolom dari table dba_objects dimana rownumnya di bawah 10000

--case 1
--no compression

82. SQL> create table test_01
as
select * from dba_objects where rownum <= 10000;

```
SQL> create table test_01
as
select * from dba_objects where rownum <= 10000;
  2    3
Table created.
```

Membuat table test_01 dimana datanya sama sepert table dba_objects dan

rownumnya dibawah 10000
83. SQL> SELECT DBMS_METADATA.GET_DDL('TABLE','TEST_01') FROM dual;

```
SQL> SELECT DBMS_METADATA.GET_DDL('TABLE','TEST_01') FROM dual;

DBMS_METADATA.GET_DDL('TABLE','TEST_01')
--------------------------------------------------------------------------------

  CREATE TABLE "SYS"."TEST_01"
   (    "OWNER" VARCHAR2(128),
        "OBJECT_NAME" VAR
```

menampilkan table yang bernama test_01 darid ual
84. SQL> ANALYZE TABLE TEST_01 COMPUTE statstcs;

```
SQL> ANALYZE TABLE TEST_01 COMPUTE statistics;

Table analyzed.
```

Melakukan analyze statsctcs table test_01
85. SQL> select blocks, pct_free , compression, compress_for
from u ser_tables
where table_name = 'TEST_01';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_01';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
       180         10 DISABLED
```

MENAMPILKAN blocks, pct_free, compression, compress_for dari table user tables dimaan table namenya adalah TEST_01
--CASE 2
--with basic compression enabled
86. SQL> create table test_02 compress basic
as
select * from dba_objects where rownum <= 10000;

```
SQL> create table test_02 compress basic
as
select * from dba_objects where rownum <= 10000;
  2    3
Table created.
```

Membuat sebuah table bernama test_02 dari data dba_objects dimana rownumnya 10000 dan mengaktfan compress tngkat basic. Sebelumnya error karena terdapat table yang bernama sama. Jadinya harus di drop terlebih dahulu dan baru membuat table nya kembali

87. SQL> ANALYZE TABLE TEST_02 COMPUTE statstcs;

```
SQL> ANALYZE TABLE TEST_02 COMPUTE statistics;

Table analyzed.
```

Melakukan analyze statstcs pada table test_02

88. SQL> select blocks, pct_free , compression, compress_for
from user_tables
where table_name = 'TEST_02';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_02';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
        37          0 ENABLED  BASIC
```

Menampilkan blocks, pct_free, compression, compress_for dari table user_tables dimana table namenya adalah TEST_02

--CASE 3

--Normal insert into empty table defned as compress basic (no efects)

89. SQL> create table test_03 compress basic
as
select * from dba_objects where 1=2;

```
SQL> create table test_03 compress basic
as
select * from dba_objects where 1=2;
  2    3
Table created.
```

Membuat table test_03 dengan data yang di copy dari dba objects yang dimana 1 adalah 2, sudah di compress

90. SQL> select * from test_03;

```
SQL> select * from test_03;

no rows selected
```

Menampilkan isi kolom dari table test_03

91. SQL> insert into test_03
select * from dba_objects where rownum <= 10000;

```
SQL> insert into test_03
select * from dba_objects where rownum <= 10000;
  2
10000 rows created.
```

Menambahkan data pada table test_03 dengan menyalin data dari dba_objects dimana rownum nya <= 10000. Sebelumnya error karena terdapat table yang bernama sama, namun di drop terlebih dahulu baru create table kembali

92. SQL > commit;

```
SQL> commit;

Commit complete.
```

Menyimpan transaksi

93. SQL> select blocks, pct_free , compression, compress_for
from user_tables
where table_name = 'TEST_03';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_03';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
                    0 ENABLED  BASIC
```

Menampilkan blocks, pct_free, compression, compress_for dari table user_tables Dimana table name nya adalahTEST_03

94. SQL> ANALYZE TABLE TEST_03 COMPUTE statstcs;

```
SQL> ANALYZE TABLE TEST_03 COMPUTE statistics;

Table analyzed.
```

Menganalisis statstcs untuk table test_03

95. SQL> select blocks, pct_free , compression, compress_for
from user_tables
where table_name = 'TEST_03';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_03';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
       162          0 ENABLED  BASIC
```

Menampilkan blocks, pct_free, compression, compress_for dari table user_tables dimaana table_name nya adalah TEST_03

--4. Direct path insert into empty table defned as compressed

96. SQL> create table TEST_04 compress basic
as
select * from dba_objects where rownum = 0;

```
SQL> create table TEST_04 compress basic
as
select * from dba_objects where rownum = 0;
  2    3
Table created.
```

Membuat table test_04 dengan data yang dimasukkan atau dicopy dari table dba_objects dimana rownumnya adalah 0, itu di compress. Sebelumnya error karena
terdapat table yang bernama sama, namun di drop terlebih dahulu baru create table kembali

97. SQL> select * from TEST_04;

```
SQL> select * from TEST_04;

no rows selected
```

Menampilkan isi kolom dari table test_04

98. SQL> insert /*+ append */ into TEST_04 select * from dba_objects where rownum <=
10000;

```
SQL> insert /*+ append */ into TEST_04 select * from dba_objects where rownum <=
 10000;

10000 rows created.
```

Melakukan penambahan data pada table test_04 dimaan mengcopy data dari dba_objects dan rownum nya adalah 10000

/*

The APPEND hint tells the optmizer to perform a direct-path insert, which improves the

performance of INSERT .. SELECT operatons for a number of reasons:

Data is appended to the end of the table, rather than atemptng to use existng free space within the table.

Data is writen directly to the data fles, by-passing the bufer cache.

*/

99. SQL> ANALYZE TABLE TEST_04 COMPUTE statstcs;

```
SQL> ANALYZE TABLE TEST_04 COMPUTE statistics;

Table analyzed.
```

Menganalisis stats table test_04

100.

SQL> select blocks, pct_free , compression, compress_for

from user_tables

where table_name = 'TEST_04';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_04';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
        37          0 ENABLED  BASIC
```

Menampilkan blocks, pct_free, compression, compress_for

Dari table user_tables dimana table name nya adalah test_04

--case 5

--without compression, then change to compressed

101.

SQL> create table test_05

as

select * from dba_objects where rownum <= 10000;

```
SQL> create table test_05
as
select * from dba_objects where rownum <= 10000;
  2    3
Table created.
```

Membuat table test_05 dimaana mengcopy dari table dba_objects dan rownumnya adalah 10000. Sebelumnya error karena terdapat table yang bernama sama, namun di drop terlebih dahulu baru create table kembali

102.

SQL> select blocks, pct_free , compression, compress_for
from user_tables
where table_name = 'TEST_05';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_05';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
                   10 DISABLED
```

Menampilkan blocks, pct_free, compression, compress_for dari table user_tables dan table namenya adalah TEST_05

103.

SQL> alter table test_05 compress basic;

```
SQL> alter table test_05 compress basic;

Table altered.
```

Mengubah table test_05 dimana ketka menambahkan data, data tersebut di compress

104.

SQL> select blocks, pct_free , compression, compress_for
from user_tables
where table_name = 'TEST_05';

```
SQL>  select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_05';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
                   10 ENABLED  BASIC
```

Menampilkan blocks, pct_free, compression, compress_for dari table user_tables dimana table name nya adalah TEST_05
105.
SQL> alter table test_05 move;

```
SQL> alter table test_05 move;

Table altered.
```

Mengubah table test_05 agar dapat dipindahkan
106.
SQL> ANALYZE TABLE TEST_05 COMPUTE statstcs;

```
SQL> ANALYZE TABLE TEST_05 COMPUTE statistics;

Table analyzed.
```

Menganalisis statstc pada table test_05
107.
SQL> select blocks, pct_free , compression, compress_for
from user_tables
where table_name = 'TEST_05';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_05';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
        37          0 ENABLED  BASIC
```

Menampilkan blocks, pct_free, compressoin, compress_for pada table user_tables dimana table name nya adalah TEST_05
==========
==========
--Advanced row compression
/*
Rows inserted with or without using direct-path insert and updated rows
are compressed using advanced row compression.
Is enabled with CREATE TABLE … ROW STORE COMPRESS ADVANCED
Is recommended for actve OLTP environments
With advanced row compression, duplicate values in the rows and columns in a data block are stored once
at the beginning of the block
note 1:

ROW STORE COMPRESS ADVANCED and COMPRESS BASIC are not supported for
tables with more than 255
note 2: You cannot drop a column from a table that is compressed for direct-load operatons,
although you can set such a column as unused.
*/
108.
SQL> create table test_06
as
select * from dba_objects where rownum <= 20000;

```
SQL> create table test_06
as
select * from dba_objects where rownum <= 20000;
  2    3
Table created.
```

Membuat table test_06 dan data dicopy dari dba_objects dimana rownum dibawah sama dengan 20000. Sebelumnya error karena terdapat table yang bernama sama, namun di drop terlebih dahulu baru create table kembali
109.
SQL> ANALYZE TABLE test_06 COMPUTE statstcs;

```
SQL> ANALYZE TABLE test_06 COMPUTE statistics;

Table analyzed.
```

Menganalisis statstcs table test_06
110.
SQL> select blocks, pct_free , compression, compress_for
from user_tables
where table_name = 'TEST_06';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_06';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
       365         10 DISABLED
```

Menampilkan blocks, pct_free, compression, compress_for dari table user_tables diamana table_name nya adalah TEST_06

111.
SQL> create table test_07 ROW STORE COMPRESS ADVANCED
as
select * from dba_objects where rownum <= 20000;

```
SQL> create table test_07 ROW STORE COMPRESS ADVANCED
as
select * from dba_objects where rownum <= 20000;
  2    3
Table created.
```

MEMBUAT table bernama test_07 dari data yang di copy dari table dba_objects dimana rownumnya 20000 dan mengaktfan kompresi baris tngkat advanced . Sebelumnya error karena terdapat table yang bernama sama, namun di drop terlebih dahulu baru create table kembali
112.
SQL> ANALYZE TABLE test_07 COMPUTE statstcs;

```
SQL> ANALYZE TABLE test_07 COMPUTE statistics;

Table analyzed.
```

Menganalisis statstcs table test_07
113.
SQL> select blocks, pct_free , compression, compress_for
from user_tables
where table_name = 'TEST_07';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_07';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
        87         10 ENABLED  ADVANCED
```

Menampilkan blocks, pct_free, compression, compress_for dari table user_tables dimana table_name nya adalah test_07

--not let us try normal insert
114.
SQL> create table test_08 ROW STORE COMPRESS ADVANCED
as
select * from dba_objects where rownum = 0;

```
SQL> create table test_08 ROW STORE COMPRESS ADVANCED
as
select * from dba_objects where rownum = 0;
  2    3
Table created.
```

Membuat table test_08 dari data yang di copy dari table dba_objects dimana rownum nya adalah 0 dan mengaktfan kompresi data tngkat advanced.
Sebelumnya error karena terdapat table yang bernama sama, namun di drop terlebih dahulu baru create table kembali

115.

SQL> select * from test_08;

```
SQL> select *from test_08
  2  ;

no rows selected
```

Menampilkan isi kolom dari table test_08

116.

SQL> insert into test_08
select * from dba_objects where rownum <= 20000;

```
SQL> insert into test_08
select * from dba_objects where rownum <= 20000;
  2
20000 rows created.
```

Memasukkan value ke dalam table test_08 dan mengcopy isi dari table dba_objects dimana rownumnya adalah 20000

117.

SQL> commit;

```
SQL> commit;

Commit complete.
```

Menyimpan transaksi

118.

SQL> ANALYZE TABLE test_08 COMPUTE statstcs;

```
SQL> ANALYZE TABLE test_08 compute statistics;

Table analyzed.
```

Melakukan analisis statstk pada table test_08

119.

SQL> select blocks, pct_free , compression, compress_for
from user_tables
where table_name = 'TEST_08';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_08';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
        92         10 ENABLED  ADVANCED
```

Menampilkan blocks, pct_free, compression, compress_for pada table user_tables dimana table_name = TEST_08

==========
==========

120.

121.

SQL> show user;
SQL> show con_name;

```
SQL> show user;
USER is "SYS"
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
```

Menampilkan user dan container yang sedang aktf sekarang

122.

SQL> CREATE TABLESPACE tbsalert
DATAFILE '/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf'
SIZE 50M LOGGING EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;

```
SQL> CREATE TABLESPACE tbsalert
DATAFILE '/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf'
SIZE 50M  LOGGING EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;
  2    3    4
Tablespace created.
```

Membuat tablespace tbsalert dengan menghubungkan tablespace ini ke fle tbsalert.dbf dan memberikan size dari fle tersebut adalah 50 mb dan setap perubahan data akan di catat ke redo log fles dan menentukan metode mengelola ruang kosong dalam blok data.

123.

SQL> select * from dba_data_fles
where TABLESPACE_NAME='TBSALERT';

```
SQL> select * from dba_data_files
where TABLESPACE_NAME='TBSALERT';
  2
FILE_NAME
--------------------------------------------------------------------------------
   FILE_ID TABLESPACE_NAME                     BYTES     BLOCKS STATUS
---------- ------------------------------ ---------- ---------- ---------
RELATIVE_FNO AUT   MAXBYTES  MAXBLOCKS INCREMENT_BY USER_BYTES USER_BLOCKS
------------ --- ---------- ---------- ------------ ---------- -----------
ONLINE_ LOST_WR
------- -------
/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf
        28 TBSALERT                         52428800       6400 AVAILABLE
        1024 NO           0          0            0   46104576        5628
ONLINE  OFF
```

Menampilkan isi kolom dari table dba_data_fles dimana tablespce name nya adalah tbsalert

124.

SQL> select * from dba_free_space
where TABLESPACE_NAME='TBSALERT';

```
SQL> select * from dba_free_space
where TABLESPACE_NAME='TBSALERT';
  2
TABLESPACE_NAME                 FILE_ID   BLOCK_ID      BYTES     BLOCKS
------------------------------ ---------- ---------- ---------- ----------
RELATIVE_FNO
------------
TBSALERT                            28        776   46071808       5624
        1024
```

Menampilkan isi kolom dari table dba_free_space dimana tablespace name nya adalah TBSALERT

125.

SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,

fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_fles df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';

```
SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,
fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_files df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';
  2    3    4    5
TABLESPACE                       FREE      BYTES   PCT_FREE
------------------------------ ---------- ---------- ----------
TBSALERT                       46071808   52428800     87.875
```

Menampilkan d ata atribute dari dba_data_fles yang di defnisikan sebagai df, dba free space didefnisikan sebagai fs dimana tablespace_name nya adalah yang sama dengan tablespace_name dan tablespace name nya adalah TBSALERT
126.
SQL> begin
DBMS_SERVER_ALERT.SET_THRESHOLD(
metrics_id => dbms_server_alert.tablespace_pct_full,
warning_operator =>DBMS_SERVER_ALERT.OPERATOR_GE,
warning_value => '55',
critcal_operator =>DBMS_SERVER_ALERT.OPERATOR_GE,
critcal_value => '70',
observaton_period => 1,
consecutve_occurrences => 1,
instance_name => 'free',
object_type =>DBMS_SERVER_ALERT.OBJECT_TYPE_TABLESPACE,
object_name => 'TBSALERT');
end;

```
SQL> begin
DBMS_SERVER_ALERT.SET_THRESHOLD(
metrics_id => dbms_server_alert.tablespace_pct_full,
warning_operator =>DBMS_SERVER_ALERT.OPERATOR_GE,
warning_value => '55',
critical_operator =>DBMS_SERVER_ALERT.OPERATOR_GE,
critical_value => '70',
observation_period => 1,
consecutive_occurrences => 1,
instance_name => 'free',
object_type =>DBMS_SERVER_ALERT.OBJECT_TYPE_TABLESPACE,
object_name => 'TBSALERT');
end;
  2    3    4    5    6    7    8    9   10   11   12   13   14  /

PL/SQL procedure successfully completed.
```

Memberikan peringatan jika tablespace bernama tbsalert sudah mencapai 55% maka
akan memberikan warning, dan jika 70% maka akan mengeluarkan peringatan critcal, di instance bernama free
127.
SQL> SELECT warning_value, critcal_value
FROM dba_thresholds
WHERE object_name='TBSALERT';

```
SQL> SELECT warning_value, critical_value
FROM dba_thresholds
WHERE object_name='TBSALERT';
  2    3
WARNING_VALUE
----------------------------------------
CRITICAL_VALUE
----------------------------------------
55
70
```

Menampilkan warning_value, critcal_value dari table dba_thresholds dimana object_name nya adalah TBSALERT
128.
SQL> SELECT * FROM dba_outstanding_alerts
WHERE object_name='TBSALERT';

```
SQL> SELECT * FROM dba_outstanding_alerts
 WHERE object_name='TBSALERT';
  2
no rows selected
```

Menampilkan isi kolom dari table dba_outstanding_alert dimana object_namennya adalah TBSALERT
129.
SQL> create table test100 (emp_id number, name varchar2(100))
tablespace TBSALERT;

```
SQL> create table test100 ( emp_id number, name varchar2(100) )
 tablespace TBSALERT;
  2
Table created.
```

Membuat table test100 dengan atribute emp_id dan tpe nya adalah number, name

tpenya varchar2 dan maksimal karakter nya adalah 100 dari tablespace TBSALERT
130.
SQL> begin
for i in 1..1000000
loop
insert into test100 values (i, 'any dummy name');
end loop;
commit;
end;

```
SQL> begin
 for i in 1..1000000
 loop
 insert into test100 values ( i, 'any dummy name' );
  2    3    4    5
  6  end loop
  7  ;
  8  commit;
  9  end;
 10  /

PL/SQL procedure successfully completed.
```

Melakukan looping dengan mamsukkan data sebanyak 1000000 pada table test100
131.
SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,
fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_fles df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';

```
SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,
fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_files df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';
  2    3    4    5
TABLESPACE                          FREE      BYTES   PCT_FREE
------------------------------ ---------- ---------- ----------
TBSALERT                           983040   52428800      1.875
TBSALERT                         15728640   52428800         30
```

Menampilkan atribute yang ada di dba_data_fles yang di defnisikan sebagai df,
dba_free_space yang didefnisikan sebagai fs dimana tablespace_name yang cocok
dengan tablespace_name dan tablespace_name nya adalah TBSALERT
--Wait a few minutes. 10-15
132.

SQL> SELECT reason, message_type , message_level
FROM dba_outstanding_alerts
WHERE object_name='TBSALERT';

```
SQL> SELECT reason, message_type , message_level
FROM dba_outstanding_alerts
WHERE object_name='TBSALERT';
  2    3
no rows selected
```

Menampilkan reason, message_type, message_level dari table dba_outstanding_alerts dimana object_name nya adalah TBSALERT

==========

==========

133.

134.

SQL> show user;

SQL> show con_name;

```
SQL> show user;
USER is "SYS"
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
```

Menampilkan user dan container name yang sedang aktf sekarang

135.

SQL> drop TABLESPACE tbsalert INCLUDING CONTENTS and DATAFILES;

```
SQL> drop TABLESPACE tbsalert INCLUDING CONTENTS and DATAFILES;

Tablespace dropped.
```

menghapus pada tablespace tbsalert termasuk konten dan dataflesnya

136.

SQL> CREATE TABLESPACE tbsalert
DATAFILE '/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf'
SIZE 50M LOGGING EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;

```
SQL> CREATE TABLESPACE tbsalert
DATAFILE '/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf'
SIZE 50M  LOGGING EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;
  2    3    4
Tablespace created.
```

Membuat tablespace tbsalert dengan menghubungkan tablespace ini ke fle tbsalert.dbf dan memberikan size dari fle tersebut adalah 50 mb dan setap perubahan data akan di catat ke redo log fles dan menentukan metode mengelola ruang kosong dalam blok data.

137.

SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes, fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_fles df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';

```
SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,
fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_files df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';
  2    3    4    5
TABLESPACE                       FREE      BYTES   PCT_FREE
------------------------------ ---------- ---------- ----------
TBSALERT                       46071808   52428800     87.875
```

Menampilkan atribute dari table dba_data fles yang didefnisikan sebagai df, dba free yang di defnisikan sebagai fs dimana tablespace_name nya cocok dengan tablespace_name dan tablespace name nya adalah TBSALERT

138.

SQL> create table test100 (emp_id number, name varchar2(100))
tablespace TBSALERT;

```
SQL> create table test100 ( emp_id number, name varchar2(100) )
 tablespace TBSALERT;
  2
Table created.
```

Membuat table baru bernama test100 dan memiliki atribute yaitu emp_id dengan tpe number, name dengan tpe varchar2 dan maksimal karakter adalah 100 dari tablespace tbsalert

139.

SQL> begin
for i in 1..1000000

loop
insert into test100 values (i, 'any dummy name');
end loop;
commit;
end;

```
SQL> begin
 for i in 1..1000000
 loop
 insert into test100 values ( i, 'any dummy name' );
  2    3    4    5
  6  end loop;
  7  commit;
  8  end;
  9  /

PL/SQL procedure successfully completed.
```

Melakukan looping yaitu memasukkan data sebanyak 1000000 ke table test100
140.
SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,
fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_fles df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';

```
SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,
fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_files df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';
  2    3    4    5
TABLESPACE                       FREE      BYTES   PCT_FREE
------------------------------ ---------- ---------- ----------
TBSALERT                         983040   52428800      1.875
TBSALERT                       15728640   52428800         30
```

Menampilkan atribute dari table dba_data_fles yang di defnisikan sebagai df, dan dba_free_space yang di defnisikan sebagai fs dimana df t ablespace_name nya adalah yang cocok dengan tablespace name dari fs dan tablespace name nya adalah
TBSALERT
141.
SQL> delete from test100;

```
SQL> delete from test100;

1000000 rows deleted.
```

Menghapus semua pada table test100

142.

SQL> commit;

```
SQL> commit;

Commit complete.
```

Menyimpan transaksi

143.

SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,
fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_fles df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';

```
SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,
fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_files df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';
  2    3    4    5
TABLESPACE                          FREE      BYTES   PCT_FREE
------------------------------ ---------- ---------- ----------
TBSALERT                           983040   52428800      1.875
TBSALERT                         15728640   52428800         30
```

Menampilkan atribute dari table dba_data_fles yang di defnisikan sebagai df, dan dba_free_space yang di defnisikan sebagai fs. dimana table space_name df cocok dengan tablespace_name fs, dan tablespace_name nya adalah TBSALERT

--Create a Segment Advisor Task

144.

SQL> DECLARE
tname VARCHAR2(128) := 'my_seg_task3';
tname_desc VARCHAR2(128) := 'Get shrink advice for segments in TBSALERT';
task_id NUMBER;
object_id NUMBER;
objectname VARCHAR2(100);
objectype VARCHAR2(100);
BEGIN
dbms_advisor.create_task('Segment Advisor', task_id,tname,tname_desc,NULL);

dbms_advisor.create_object(tname,'TABLESPACE','TBSALERT',' ',' ',NULL, '
',object_id) ;
dbms_advisor.set_task_parameter(tname,'RECOMMEND_ALL','TRUE');
END;

```
 objecttype VARCHAR2(100);
BEGIN
 dbms_advisor.create_task('Segment Advisor', task_id,tname,tname_desc,NULL);
 dbms_advisor.create_object(tname,'TABLESPACE','TBSALERT',' ',' ',NULL, ' ',obje
ct_id) ;
 dbms_advisor.set_task_parameter(tname,'RECOMMEND_ALL','TRUE');
 END;
  2    3    4    5    6    7    8    9   10   11   12   13  /
DECLARE
*
ERROR at line 1:
ORA-13607: The specified task or object my_seg_task3 already exists
ORA-06512: at "SYS.PRVT_ADVISOR", line 6181
ORA-06512: at "SYS.PRVT_ADVISOR", line 1851
ORA-06512: at "SYS.DBMS_SYS_ERROR", line 86
ORA-06512: at "SYS.PRVT_ADVISOR", line 7234
ORA-06512: at "SYS.PRVT_ADVISOR", line 1624
ORA-06512: at "SYS.PRVT_ADVISOR", line 6143
ORA-06512: at "SYS.DBMS_ADVISOR", line 119
ORA-06512: at line 9
Help: https://docs.oracle.com/error-help/db/ora-13607/
```

Membuat dan melakukan konfgurasi task menggunakan utlitas dbms_advisor, tapi error karena task yang dibuat sudah ada di database
--Execute the task.
145.
SQL> DECLARE
tname VARCHAR2(128) := 'my_seg_task3';
BEGIN
dbms_advisor.EXECUTE_TASK(tname);
END;

```
SQL> DECLARE
tname VARCHAR2(128) := 'my_seg_task3';
BEGIN
dbms_advisor.EXECUTE_TASK(tname);
END;
  2    3    4    5    6  /
DECLARE
*
ERROR at line 1:
ORA-13630: The task my_seg_task3 contains execution results and cannot be
executed.
ORA-06512: at "SYS.DBMS_ADVISOR", line 259
ORA-06512: at "SYS.PRVT_ADVISOR", line 3594
ORA-06512: at "SYS.DBMS_SYS_ERROR", line 86
ORA-06512: at "SYS.PRVT_ADVISOR", line 3341
ORA-06512: at "SYS.DBMS_ADVISOR", line 309
ORA-06512: at "SYS.DBMS_ADVISOR", line 254
ORA-06512: at line 4
Help: https://docs.oracle.com/error-help/db/ora-13630/
```

Melakukan atau menjalankan tugas analisis yang akan menganalisis tablespace TBSALERT. Tapi error karena sama sepert di atas

146.

SQL> SELECT DESCRIPTION FROM dba_advisor_tasks WHERE TASK_NAME='my_seg_task3';

```
SQL> SELECT DESCRIPTION FROM dba_advisor_tasks WHERE TASK_NAME='my_seg_task3';

DESCRIPTION
-------------------------------------------------------------------------------
Get shrink advice for segments in TBSALERT
```

Menampilkan descripton dari table dba_advisor tasks dimaan task name nya adalah my seg task3

147.

SQL> SELECT atr1, atr2, message
FROM dba_advisor_fndings f, dba_advisor_objects o
WHERE f.task_name = o.task_name
AND f.object_id = o.object_id
AND f.task_name = 'my_seg_task3';

```
SQL> SELECT DESCRIPTION FROM dba_advisor_tasks WHERE TASK_NAME='my_seg_task3';

DESCRIPTION
--------------------------------------------------------------------------------
Get shrink advice for segments in TBSALERT

SQL>  SELECT attr1, attr2, message
FROM dba_advisor_findings f, dba_advisor_objects o
 WHERE f.task_name = o.task_name
 AND f.object_id = o.object_id
 AND f.task_name = 'my_seg_task3';
  2    3    4    5
ATTR1
--------------------------------------------------------------------------------
ATTR2
--------------------------------------------------------------------------------
MESSAGE
--------------------------------------------------------------------------------
SYS
TEST100
Enable row movement of the table SYS.TEST100 and perform shrink, estimated savin
gs is 29351936 bytes.
```

menampilkan atre1, atr2, message dari table dba_advisor_fndings di defnisikan seabgai f, dba_advisor_objects di defnisikan sebagai o dimana task_name dari f adalah cocok dengan task_name o, dan object_id dari f cocok dengan object_id nya o, dan task_name f adalah my_seg_task3

148.

SQL> alter table test100 enable row movement;

```
SQL> alter table test100 enable row movement;

Table altered.
```

Mengubah untuk mengaktfan row movement pada table test100

149.

SQL> ALTER TABLE test100 SHRINK SPACE;

```
SQL> ALTER TABLE test100 SHRINK SPACE;

Table altered.
```

Mengubah atau melakukan pemulihan ruang disk.

150.

SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,
fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_fles df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';

```
SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,
fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_files df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';
  2    3    4    5
TABLESPACE                       FREE      BYTES   PCT_FREE
------------------------------ ---------- ---------- ----------
TBSALERT                       46006272   52428800      87.75
```

Menampilkan atribute dari table dba_data_fles yang di defnisikan sebagai df, dba_free_space yang di defnisikan sebagai fs dimana tablespace name df cocok dengan tablespace_name dari fs. dan tablespace name nya df adalah TBSALERT

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/*

Managing Resumable Space Allocaton

A resumable statement:

1- Enables you to suspend large operatons instead of receiving an error

2- Gives you a chance to fx the problem while the operaton is suspended, rather than startng over

3-Is suspended for the following conditons:

>>Out of space

>>Maximum extents reached

>>Space quota exceeded

4-Can be suspended and resumed multple tmes

*/

151.

152.

SQL> show user;

SQL> show con_name;

```
SQL> show user
USER is "SYS"
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
```

Menampilkan user dan container name yang aktf sekarang

153.

SQL> drop TABLESPACE tbsalert INCLUDING CONTENTS and DATAFILES;

```
SQL> drop TABLESPACE tbsalert INCLUDING CONTENTS and DATAFILES;

Tablespace dropped.
```

menghapus tablespace nya tbsalert ternasuk contents dan datafles
154.
SQL> CREATE TABLESPACE tbsalert
DATAFILE '/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf'
SIZE 10M LOGGING EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;

```
SQL> CREATE TABLESPACE tbsalert
DATAFILE '/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf'
SIZE 10M  LOGGING EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;
  2    3    4
Tablespace created.
```

Membuat tablespace tbsalert dengan menghubungkan tablespace ini ke fle tbsalert.dbf dan memberikan size dari fle tersebut adalah 10 mb dan setap perubahan data akan di catat ke redo log fles dan menentukan metode mengelola ruang kosong dalam blok data.
155.
SQL> create table test100 (emp_id number, name varchar2(100))
tablespace TBSALERT;

```
SQL> create table test100 ( emp_id number, name varchar2(100) )
 tablespace TBSALERT;
  2
Table created.
```

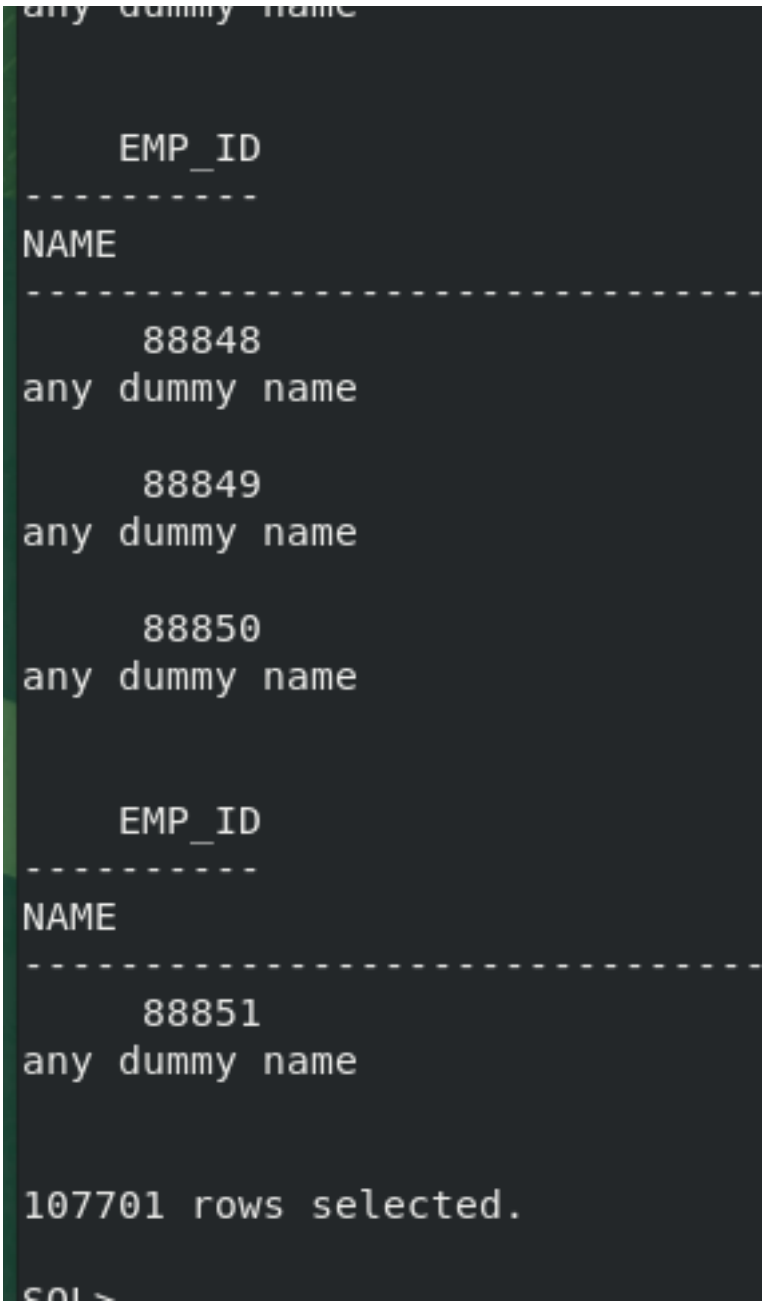
Membuat table test100 dengan atribute nya emp_id dan tpenya number, name tpenya varchar2 dan maksimal karakter 100 dari tablespace tbsalert
SQL> begin
for i in 1..400000
loop
insert into test100 values (i, 'any dummy name');
commit;
end loop;
end;

```
SQL> begin
 for i in 1..400000
 loop
 insert into test100 values ( i, 'any dummy name' );
  2    3    4    5
  6  commit;
  7  end loop;
  8  end;
  9  /
begin
*
ERROR at line 1:
ORA-01653: unable to increase tablespace TBSALERT by 1MB during insert or
update on table SYS.TEST100
ORA-06512: at line 4
Help: https://docs.oracle.com/error-help/db/ora-01653/
```

Melakukan looping sebanyak 400000 untuk memasukkan data pada table test100.tp gagal karena size nya terlalu kecil atau meminta di tambahkan

156.

SQL> select * from test100;



Menampilakn isi kolom dari table test100

/*

A statement executes in resumable mode only if its session has been enabledby one of the following actons:

1- The RESUMABLE_TIMEOUT initializaton parameter is set to a nonzero value.

2- An ALTER SESSION ENABLE RESUMABLE statement is issued

*/

157.

SQL> ALTER SESSION ENABLE RESUMABLE;

```
SQL> ALTER SESSION ENABLE RESUMABLE;

Session altered.
```

Mengaktfan ftur resumable yaitu tetap melanjutkan operasi dan mencegah error ketka terjadi masalah ruang penyimpanan

158.

SQL> truncate table test100;

```
SQL> truncate table test100;

Table truncated.
```

Menghapus seluruh isi baris dari test100

159.

SQL> begin
for i in 1..400000
loop
insert into test100 values (i, 'any dummy name');
commit;
end loop;
end;

```
SQL> begin
 for i in 1..400000
 loop
 insert into test100 values ( i, 'any dummy name' );
  2    3    4    5  commit;
  6  end loop;
  7  end;
  8  /
```

Melakukan looping sebanyak 400000 untuk memasukkan data test100, tapi disini ngestuck ga gerak uda lama, jd tdak bisa tamipilin hasil

--the script will be suspend 7,200 seconds=2 hours. (this the default)
--you can do ALTER SESSION ENABLE RESUMABLE TIMEOUT 3600
--now oracle will not give us any message that this script has been suspend
--but we can see it as DBA. open new session and do this
/*

160.

SQL> select * from dba_resumable;

```
---------- ---------- ---------- ----------------- ----------------- -----
   TIMEOUT START_TIME           SUSPEND_TIME         RESUME_TIME
---------- -------------------- -------------------- --------------------
NAME
--------------------------------------------------------------------------------
SQL_TEXT
--------------------------------------------------------------------------------
ERROR_NUMBER
------------
ERROR_PARAMETER1
--------------------------------------------------------------------------------
ERROR_PARAMETER2
--------------------------------------------------------------------------------
ERROR_PARAMETER3
--------------------------------------------------------------------------------
ERROR_PARAMETER4
--------------------------------------------------------------------------------
ERROR_PARAMETER5
--------------------------------------------------------------------------------
ERROR_MSG
--------------------------------------------------------------------------------
```

Menampilkan isi kolom dari table dba_resumable

161.

SQL> SELECT fle_name, autoextensible FROM dba_data_fles WHERE tablespace_name='TBSALERT';

```
SQL> SELECT file_name, autoextensible FROM dba_data_files WHERE
tablespace_name='TBSALERT';
  2
FILE_NAME
--------------------------------------------------------------------------------
AUT
---
/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf
NO
```

Menampilkan fle_name, autoextensible dari table dba_data_fles dimana tablespace namenya adalah TBSALERT

162.

SQL> ALTER DATABASE DATAFILE '/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf' AUTOEXTEND ON MAXSIZE unlimited;

```
SQL> ALTER DATABASE DATAFILE
'/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf'
AUTOEXTEND ON MAXSIZE unlimited;
  2    3
Database altered.
```

Mengubah databasefle datafle nya yaitu mengubah pengaturan dari fle fsik nya yaitu tbsalert.dbf dan memberikan otomats jika terjadi permasalahn dengan penyimpanan, akan menambah otomats

163.

SQL> SELECT fle_name, autoextensible FROM dba_data_fles WHERE tablespace_name='TBSALERT';

```
SQL> SELECT file_name, autoextensible FROM dba_data_files WHERE
tablespace_name='TBSALERT';
  2
FILE_NAME
--------------------------------------------------------------------------------
AUT
---
/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf
YES
```

Menampilkan fle_name autoextensible dari dba_data_fles dimana tablespace_name nya adalah TBSALERT

--WAIT untl NO ROWS retrived by THIS QUERY

164.

SQL> select * from dba_resumable;

```
--------- ------------------ ------------------- ------------------
NAME
--------------------------------------------------------------------------------
SQL_TEXT
--------------------------------------------------------------------------------
ERROR_NUMBER
------------
ERROR_PARAMETER1
--------------------------------------------------------------------------------
ERROR_PARAMETER2
--------------------------------------------------------------------------------
ERROR_PARAMETER3
--------------------------------------------------------------------------------
ERROR_PARAMETER4
--------------------------------------------------------------------------------
ERROR_PARAMETER5
--------------------------------------------------------------------------------
ERROR_MSG
--------------------------------------------------------------------------------
```

Menampilkan isi kolom dari table dba_resumable
SQL> select count(1) from test100;
165.

```
SQL> select count(1) from test100;

  COUNT(1)
----------
    107701
```

Menampilkan dan menghitung total baris data yang bernama table test100